

SEWER NETWORK TELEMETRY

Hawk

The battery powered 4G logger built for Australian sewer networks.

The Hawk installs inside the maintenance hole. It powers and reads your flow, level and quality sensors over RS485, logs as often as every second, stores every reading on board, and reports over 4G LTE to FlowSense and to your own systems. There is no cabinet to build, no power to bring to site, and no need to enter the chamber to commission or reconfigure it.

No mains

Internal battery up to 152 Ah, 40 μ A asleep.

Submersible

Sealed to IP68; keeps logging underwater.

Surface set up

Magnet wake and EMS-Flow from the road.



4G LTE	1 second	3 $\times$ RS485	152 Ah	IP68
CELLULAR TELEMETRY	FASTEST LOGGING	SENSOR PORTS	LARGEST BATTERY	SUBMERSIBLE

Where the Hawk is used

Flow surveys and model calibration

Temporary and permanent gauging to calibrate hydraulic models and plan upgrades.

Inflow and infiltration

Wet weather response across a catchment against its dry weather baseline.

Overflow structures

Level and spill detection at overflow and relief structures, for licence reporting.

Pumping stations and trunk mains

Wet well level and inflow where there is no RTU or spare capacity.

PART OF THE EDS SEWER MONITORING RANGE

One logger for every EDS sensor

Hawk connects directly to E-Flow 3 and AVFS and reports to FlowSense: one team for the sensor, logger, telemetry and data.

E-Flow 3

Dual mode sewer flowmeter

AVFS

Area velocity flow sensor

FlowSense

Hosted telemetry platform

01 Built for the maintenance hole

A sewer maintenance hole is one of the hardest places to keep electronics working. It is humid and corrosive, it can surge in wet weather, it has no power supply, and every entry is a confined space entry. Each part of the Hawk answers one of those conditions.



SURCHARGE

Sealed to IP68

Canister, lid and connectors are sealed for submersion. The Hawk keeps logging while the chamber is surcharged.



CORROSIVE ATMOSPHERE

Corrosion resistant

ABS canister, sealed connectors and a stainless wall bracket, chosen for the sewer atmosphere.



NO POWER AT SITE

Internal battery

72 to 152 Ah on board and a 40 μ A sleep current. No mains, no solar, no roadside cabinet.



CONFINED SPACE

Set up from the surface

Magnetic wake and EMS-Flow set up from the road. Configuration and firmware follow over 4G.



LONG CABLE RUNS

15 kV ESD protection

Every RS485 port is protected against the transients picked up on long sensor cables.



PATCHY COVERAGE

Every reading kept

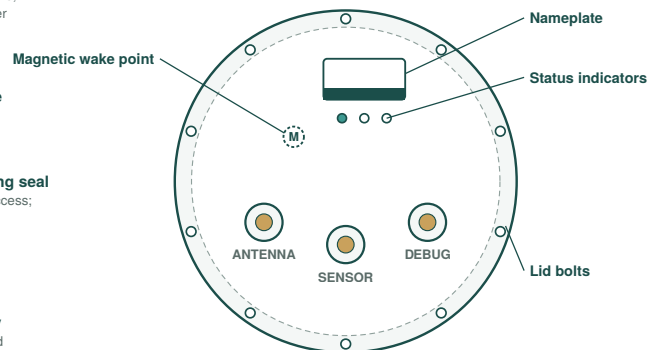
32 MB on board holds 10 years of data, so a missed report does not mean lost data.

02 Anatomy

A THE HAWK



B TOP PLATE, FROM ABOVE

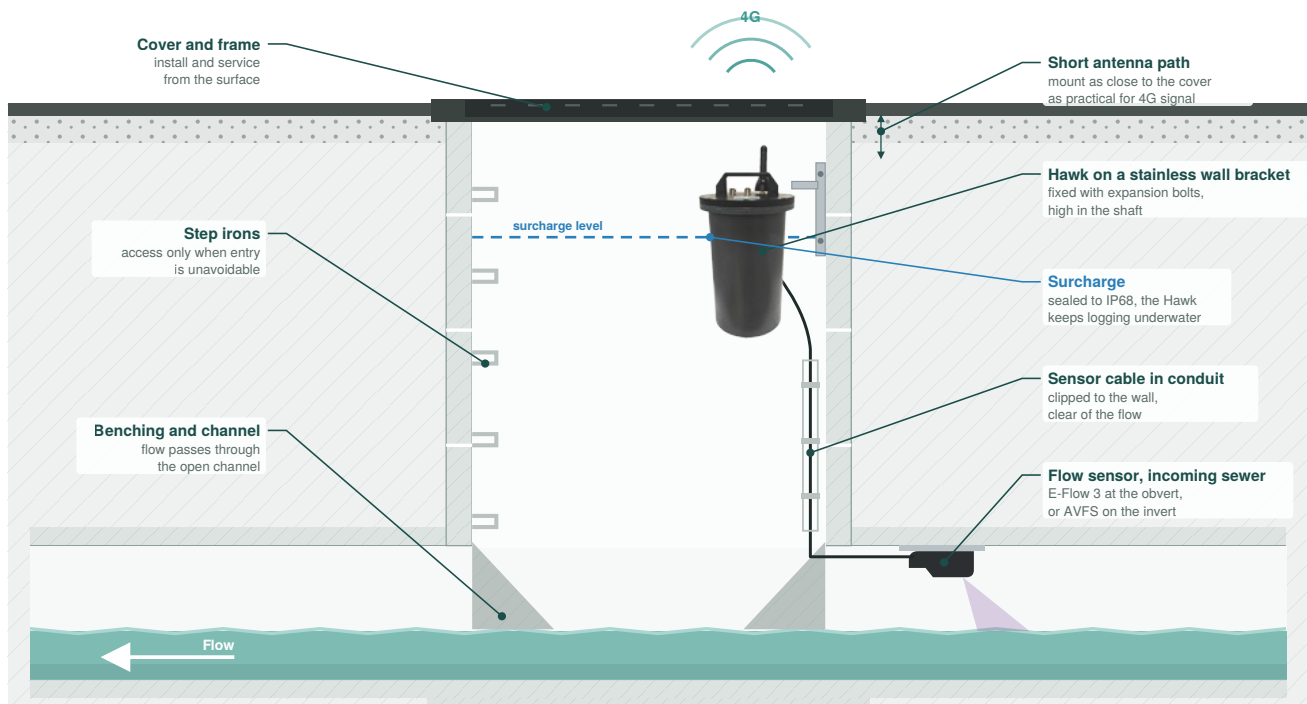


Top plate layout indicative.

Figure 1. The Hawk, and its top plate seen from above. There are no switches, buttons or external batteries to fail in the chamber.

03 Installation

The Hawk is fixed high in the maintenance hole shaft on a stainless wall bracket, as close to the cover as practical so the 4G signal has the shortest path out. Sensor cables run down the wall in conduit to the flow sensor in the sewer. Nothing is mounted at the surface, so there is nothing to damage, vandalise or maintain in the road reserve.



Schematic, not to scale. Treat every maintenance hole as a confined space.

Figure 2. A typical installation. The Hawk sits below the cover on the shaft wall with its sensor in the incoming sewer.

Location

Mount on the shaft wall clear of the step irons and the cover opening, antenna up, as high as the cover and frame allow.

Fixing

Stainless wall bracket with expansion bolts into the shaft wall. The canister hangs below the bracket and lifts out by its handle.

Safety

Treat every maintenance hole as a confined space and confirm the site hazardous area classification before installation.

04 Commissioning from the surface

1

Fit and seal

SIM fitted and the lid bolted down on its O ring seal, in the workshop or at the truck.

2

Mount and connect

Bracket fixed below the cover and sensors connected on RS485, working from the surface.

3

Wake and configure

A magnet wakes the Hawk; EMS-Flow sets it up over Bluetooth from the road.

4

Confirm and leave

The first report is confirmed in FlowSense and the cover is replaced.

No confined space entry is needed to commission, reconfigure or update the Hawk.

05 Technical specifications

TELEMETRY

Cellular	4G LTE
Logging interval	1 second to 24 hours
Reporting interval	1 minute to 24 hours
Data centres	Up to 4, each with primary and standby addressing
Configuration	Fully configurable in EMS-Flow: Bluetooth on site, remote over 4G
Firmware	Remote update over 4G
Operating modes	Low power and debug, switchable

SENSOR INTERFACE

Ports	3 × RS485, independent channels
Sensors	Modbus RTU sensors, including E-Flow 3 and AVFS
Serial settings	Baud rate, parity and stop bits set per channel
Protection	15 kV ESD on every RS485 port
Sensor power	2 × switched 12 V outputs, 500 mA (6 W) each
Warm up	Power on delay up to 255 s before each reading
Scheduling	Logging interval, warm up and retries set per channel

DATA

On board memory	32 MB, 10 years of data retention
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SENSOR CONNECTOR WIRING

 Red	12 V switched	sensor supply
 Black	0 V	power ground
 Green or blue	RS485 A	data
 Yellow	RS485 B	data

POWER

Battery	Internal, branded cells: 72, 80 or 100 Ah rechargeable, or 152 Ah primary
Operating current	Under 65 mA average at 7.2 VDC
Sleep current	40 µA or less at 7.2 VDC
External supply	None required

ENCLOSURE

Protection	IP68, submersible
Material	V0 grade flame retardant, corrosion resistant ABS
Lid	Bolted, O ring sealed; SIM and battery inside
Controls	Magnetic wake point and status indicators; no switches
Dimensions	Ø165 × 243 mm, excluding handle and antenna
Mounting	Stainless wall bracket, expansion bolts

ENVIRONMENT

Operating temperature	–40 to 60 °C
Storage temperature	–40 to 85 °C
Humidity	95 %, non condensing

OPTIONS

Chamber sensing	Methane, temperature and humidity, and tilt to flag a disturbed or dislodged unit
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Specifications are typical. Battery life depends on the logging and reporting intervals, the connected sensors and the site temperature; EDS sizes the battery for each deployment. Confirm the configuration for your site with EDS before ordering.

06 System and telemetry

The Hawk powers each sensor only while it reads it, logs as often as every second and reports over 4G LTE as often as every minute, or as seldom as once a day. It can deliver the same data to FlowSense and to a utility SCADA or historian at once, or treat them as primary and standby, so the record continues through a platform outage.

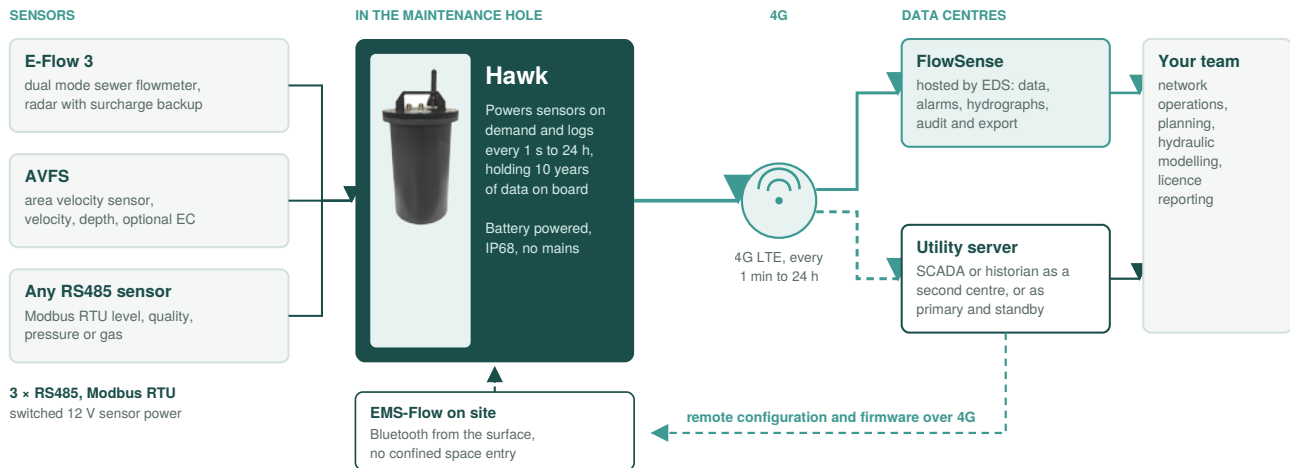


Figure 3. Sensors to the Hawk over RS485, the Hawk to your data centres over 4G.

07 Power and battery

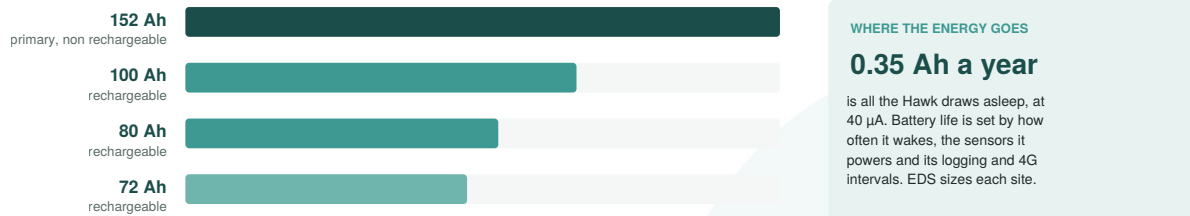
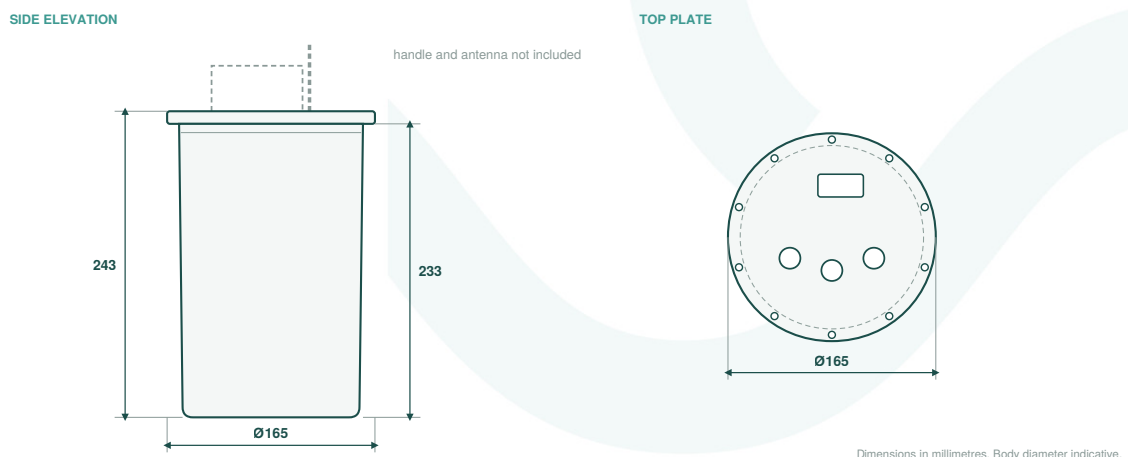


Figure 4. Battery options. The rechargeable cells suit sites that are revisited; the 152 Ah primary cell gives the longest unattended service.

08 Dimensions



09 Ordering

MODEL	BATTERY	BEST SUITED TO
Hawk 72	72 Ah rechargeable	Short surveys and frequent reporting on sites that are visited regularly
Hawk 80	80 Ah rechargeable	General network monitoring with scheduled maintenance visits
Hawk 100	100 Ah rechargeable	Long interval sites and multiple sensors on one logger
Hawk 152	152 Ah primary	The longest unattended service, for permanent and remote sites
Chamber sensing	Option	Methane, temperature and humidity, and tilt, fitted to any model

In the box

- Hawk logger
- 4G antenna
- User manual
- Product certificate

Accessories

- Stainless wall bracket and expansion bolts
- RS485 sensor cable to length
- RS485 debug cable
- SIM and data plan

EDS services

- Site assessment and battery sizing
- Installation and commissioning
- FlowSense hosting and alarms
- Maintenance and data services

Works with FlowSense

Pair the Hawk with FlowSense, the EDS telemetry platform, for hosted data, hydrographs, alarms, data quality auditing and export to hydraulic modelling and corporate systems. Hawk can report to your own server at the same time, so your data stays in your hands.

DOCUMENT HAWK DS 001	REVISION A	ISSUED September 2026	STATUS Released	ISSUED BY EDS, Brisbane
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TALK TO EDS ABOUT THE HAWK

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Environmental Data Services

13/20 to 22 Ellerslie Road, Meadowbrook QLD 4131

Tell us the network, the sites and the data you need. EDS will size the battery, supply the bracket and sensors, install and commission from the surface, and deliver the data through FlowSense. Hawk is also available with EDS rental fleets for temporary flow surveys.

Australian owned and working in water and wastewater since 1991, with field crews across Australia.